

# Software Engineering COMP1035

## Lecture 08

### *Prototyping*

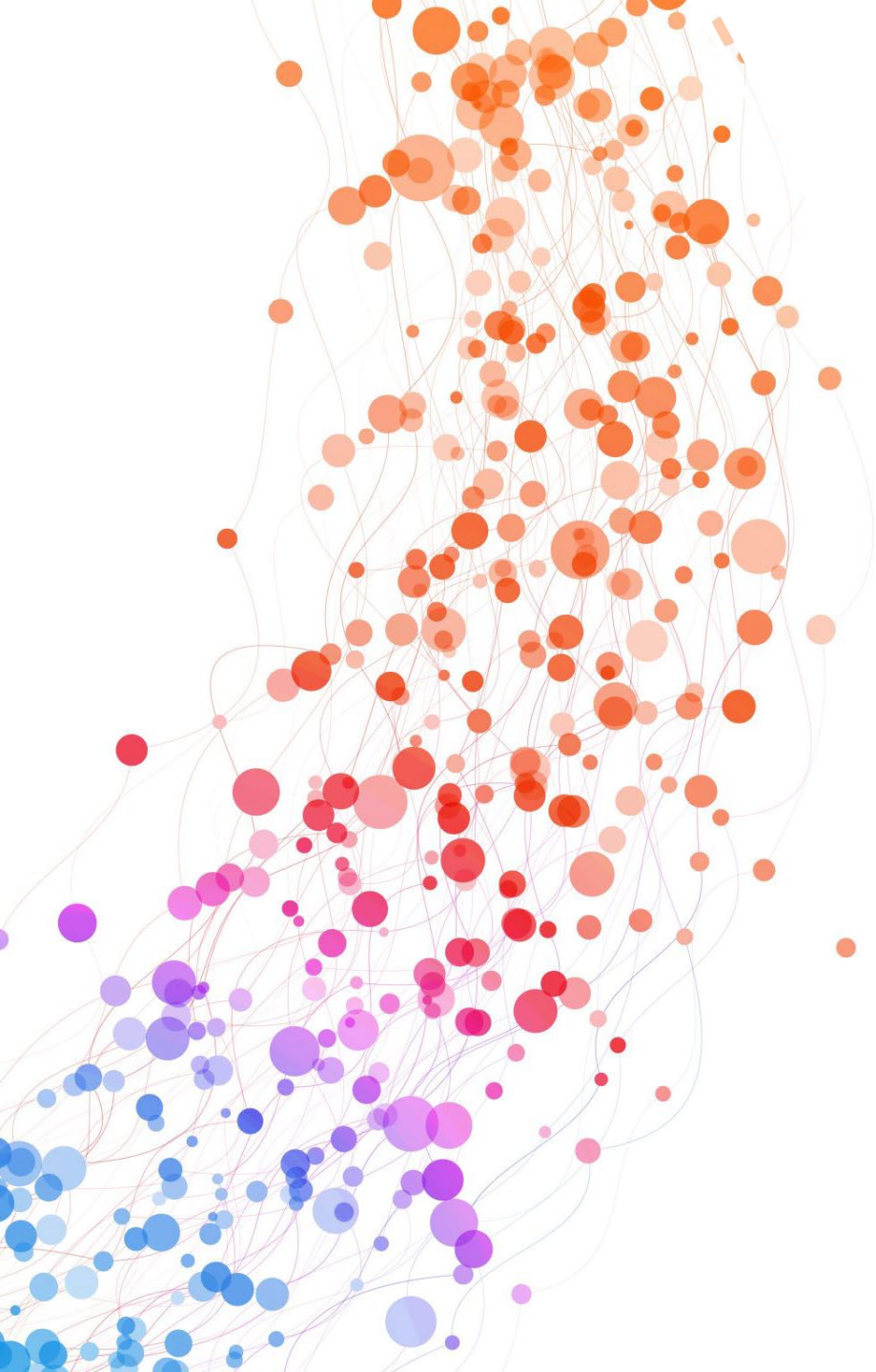




# Today's Learning Outcomes

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1. Prototypes are high level designs to help to develop system requirements/specifications.
2. There are different types of prototypes – to test different ideas.
3. They are good for “communicating” and checking those ideas.
4. Prototyping risks.
5. Prototyping tools.



# What are Prototypes?

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# System Requirements >> Prototypes

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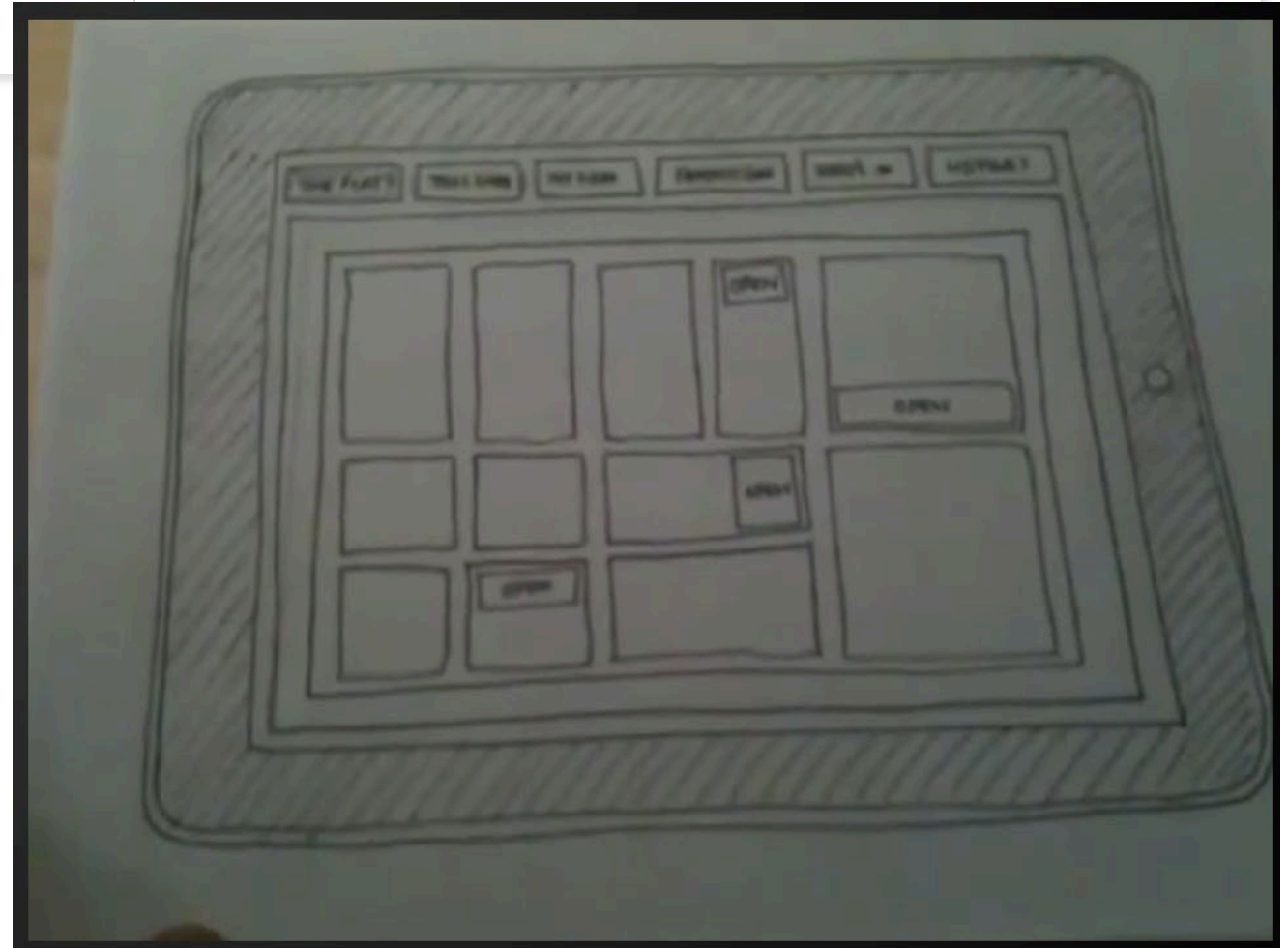
- System requirements/specifications are
  - Good for a **checklist of things to achieve**.
  - Hard to understand (unless you wrote them) the overall idea.
  - Often full of conflicting specifications.
  - Bad for conveying the overall idea.
- Prototypes are
  - A way of **envisioning how all your specifications work together**.
  - A way of testing the consistency of your specifications.
  - Easy to show people and talk about.

# System Requirements >> Prototypes

## 3.1 Capability or Functional Requirements

The requirements associated with the functionality of the APAF ground data system are itemized in the table below. The mnemonic, *FR*, is used in the requirement identifiers to show that these requirements are Functional Requirements unless otherwise specified.

| Requirement Identifier | Requirement Description                                                                                                                                                     | Verification Method(s)     | Source                                                     | Rationale & Comments                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------|
| APAF-FR-01             | The APAF system shall acquire from ESOC the telemetry data of the ASPERA-3 Experiment and Mars Express Orbit/Attitude.                                                      | Demonstration              | ESOC Data Disposition system Interface Document            | To automatically process the data on a daily basis.                                   |
| APAF-FR-02             | The APAF system shall process all ASPERA-3 science data into IDFS data sets.                                                                                                | Demonstration and Analysis | Proposal, NASA Discovery Office Program-Level Requirements | Section 4.1.1 of Program-Level Requirements, Baseline Science Requirements. APAF PDMP |
| APAF-FR-03             | The APAF system shall process the engineering and ancillary information necessary for calibration and science validation into IDFS data sets.                               | Demonstration and Analysis | NASA Discovery Office Program-Level Requirements           | Section 4.5.1, Science Data Management                                                |
| APAF-FR-04             | Intermediate files of cleaned-up ASPERA-3 and MEX OA telemetry shall be generated in the event that cleaned-up telemetry is not provided by ESOC.                           | Demonstration and Analysis | APAF Software Requirements Review Minutes                  | To support the ASPERA-3 team in meeting mission goals and objectives                  |
| APAF-FR-05             | The ASPERA-3 and MEX OA telemetry data shall be stored on a local SwRI archive.                                                                                             | Demonstration              | APAF Project Data Management Plan                          | For data availability and re-processing                                               |
| APAF-FR-06             | The ASPERA-3 and MEX OA IDFS data sets shall be stored on a local SwRI archive.                                                                                             | Demonstration              | APAF Project Data Management Plan                          | For data availability and analysis                                                    |
| APAF-FR-06a            | Any APAF-generated intermediate files of ASPERA-3 and MEX OA cleaned-up telemetry shall be stored on a local SwRI archive.                                                  | Demonstration              | APAF Software Requirements Review Minutes                  | For data availability and re-processing, and to support the ASPERA-3 team             |
| APAF-FR-07             | Web-based displays of the most current ASPERA-3 data shall be provided for public view.                                                                                     | Demonstration              | APAF Project Data Management Plan                          | E/PO and monitor instrument performance                                               |
| APAF-FR-08             | Web-based displays defined by the ASPERA-3 team shall be provided where any available ASPERA-3 data (as opposed to just the most current) can be used for science analysis. | Demonstration              | APAF Software Requirements Review Minutes                  | To support the ASPERA-3 team in meeting mission goals and objectives                  |





# What are Prototypes?

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- A prototype is a concrete but partial representation or [partial] implementation of a system.

”System Prototyping, where a version of the system or **part of the system is developed quickly** to check the customer’s requirements and the feasibility of some design decisions”

- Prototypes are used extensively in most design and construction domains.

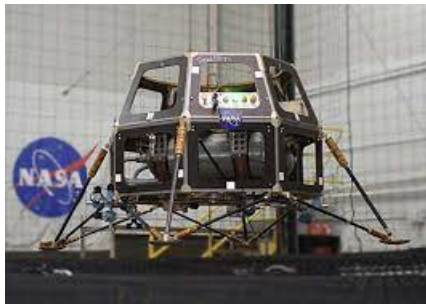


# What are Prototypes?

- Can be used in a software development process to help anticipate changes:
  - **Requirement engineering process** – help with elicitation and validation of system requirements.
  - **System design process** - used to explore software solutions and in the development of a user interface for the system.

# Prototyping in Engineering

- Nasa engineers built full-size prototypes of moon lander.
- Check visibility through windows, control etc.
  - Whilst reducing weight ratio.
- Decided that astronauts should stand and not sit, to reduce window size and save overall weight.



## TM-1

Our next mockup was a full-sized wooden model of the complete LM, descent and ascent stages, containing as much engineering detail as we could get into it before the review in March 1964. Its focus was on the crew compartment—especially the support and restraint, displays and controls, equipment stowage, and lighting—and on egress to the lunar surface. We were able to include realistic mockups of some equipment, such as the ascent and descent, engines, environmental control system components, and radar and communications antennas. Working models of the hand controllers with which the astronauts would fly the LM were provided at both pilot stations, allowing the crew to experience the tactile feel of the controls as they stood in flight position hooked into the support and restraint devices.

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## M-1

As our ideas took shape for the standup crew position and cylindrical flat-faced crew compartment with canted triangular windows, we checked out their feasibility in a simple wood-and-foam-board mockup of the forward interior portion of the cockpit. We converted this mockup into drawings and sketches from which the more complete M-1 could be made, adding the tanks, rocket engine bell, electronic equipment bay, antennas, and other external features of the LM ascent stage. Throughout spring and early summer 1963 the engineering design groups added to this mockup design definition. A formal review of M-1 was scheduled with NASA for mid-September.





# Low vs High Fidelity





Low vs High Fidelity

# Low vs High Fidelity



# Low vs High Fidelity

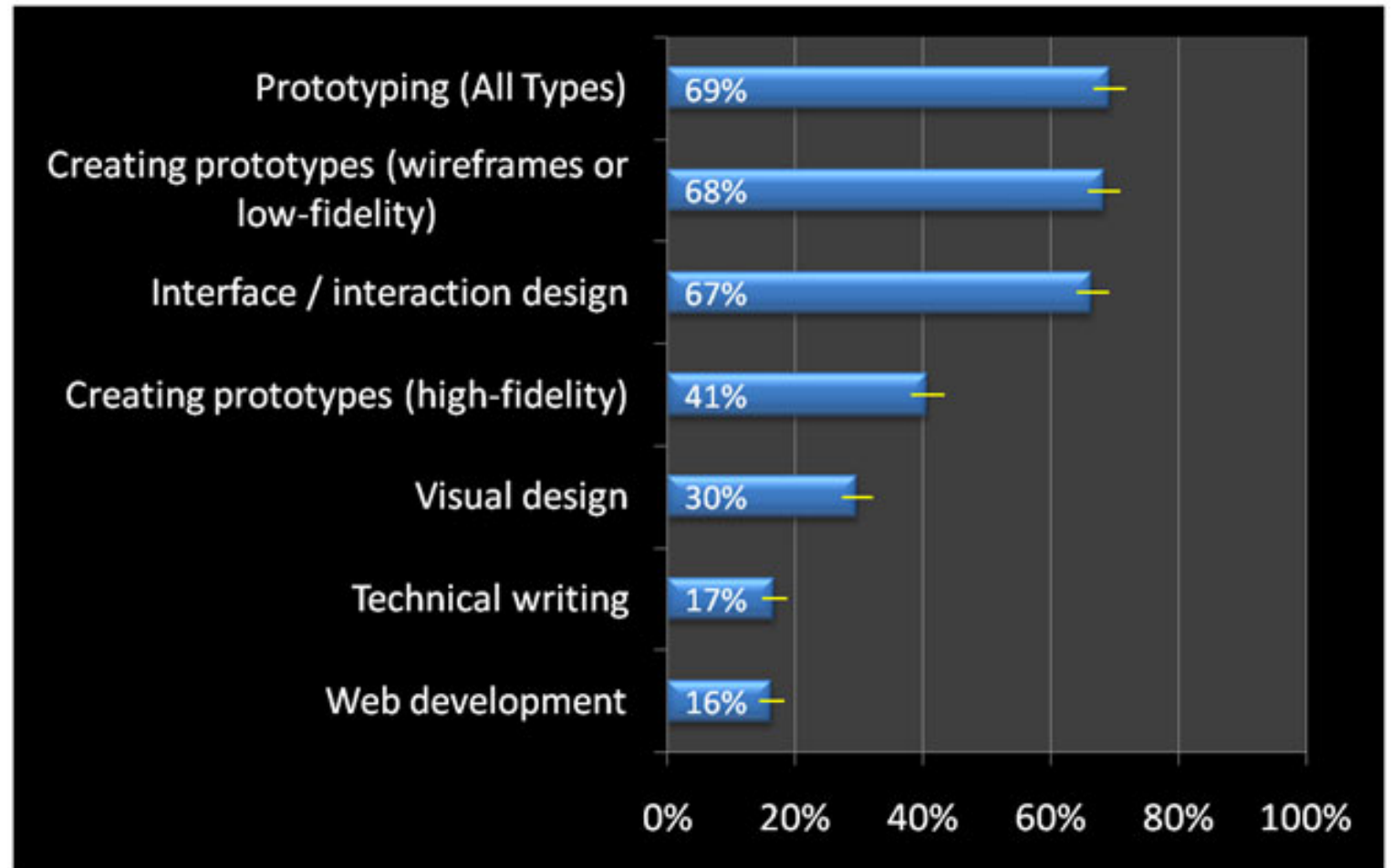




# Low vs High Fidelity

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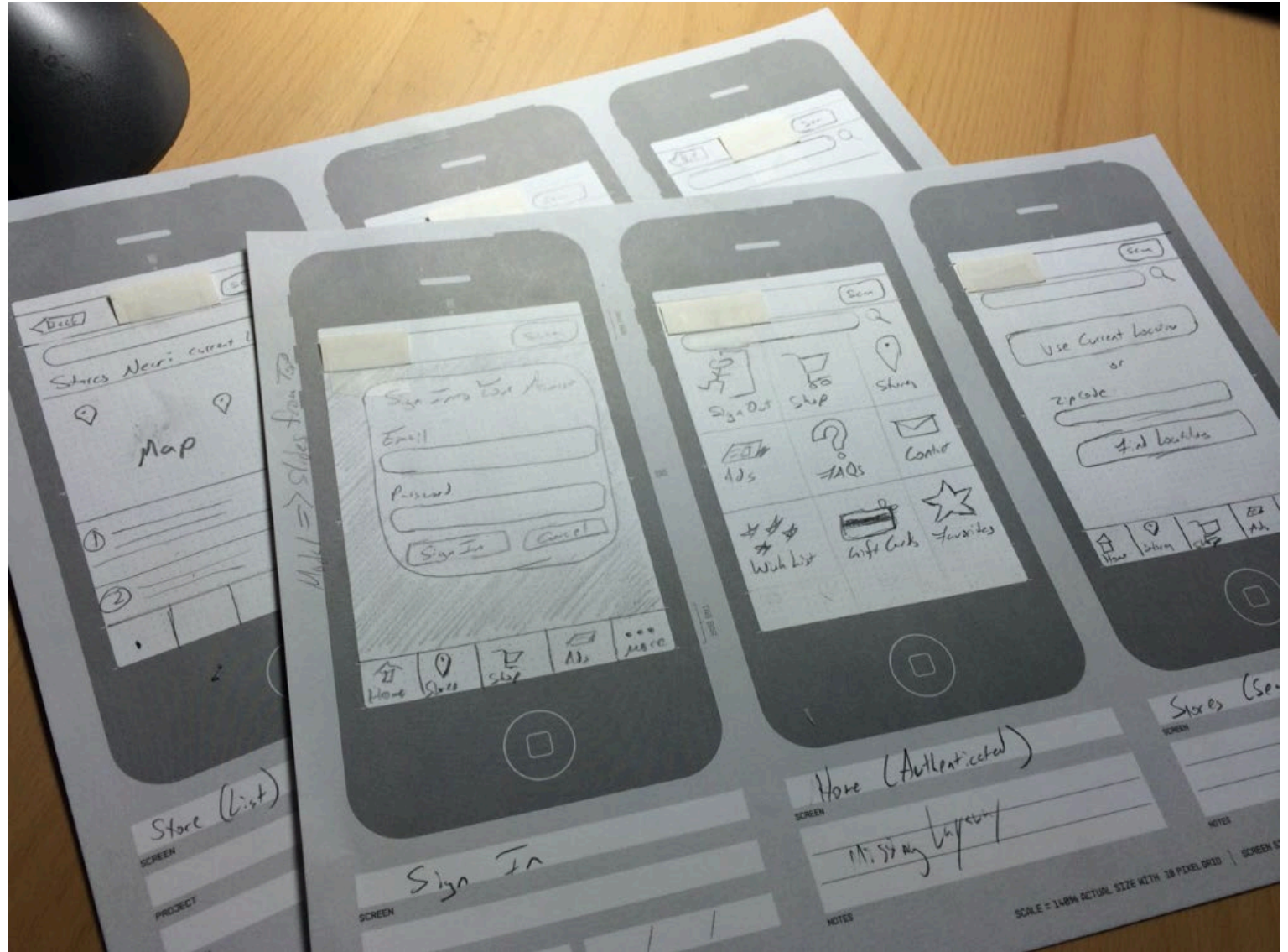
<https://measuringu.com/ux-methods/>





# Low Fidelity: Sketching

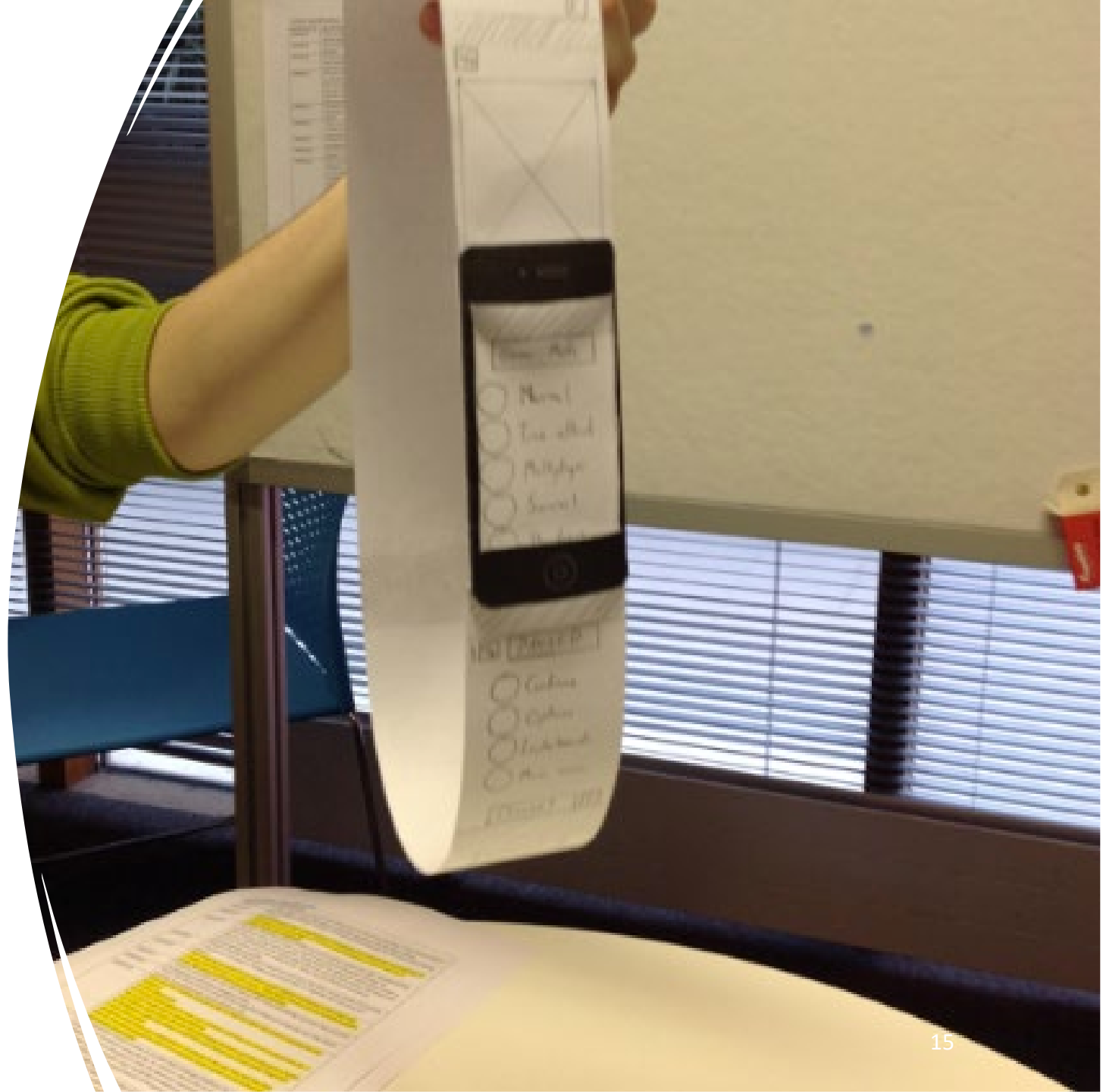
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# Low Fidelity: Sketching

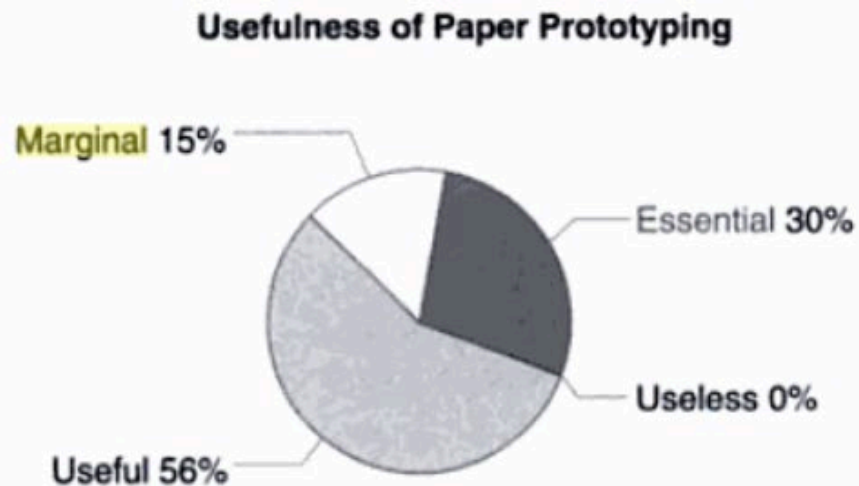
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Paper prototype by UNUK  
student X years ago.

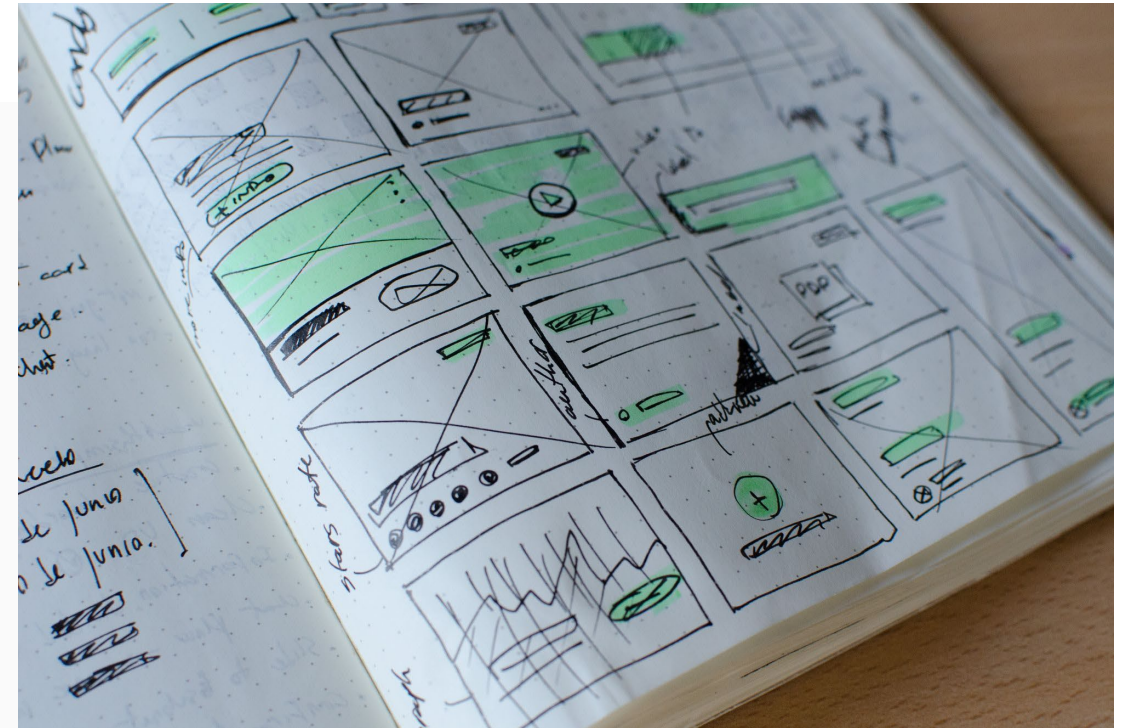


# Low Fidelity Prototyping

Snyder 2003 – Paper Prototyping – the fast and easy way to design and refine user interfaces.



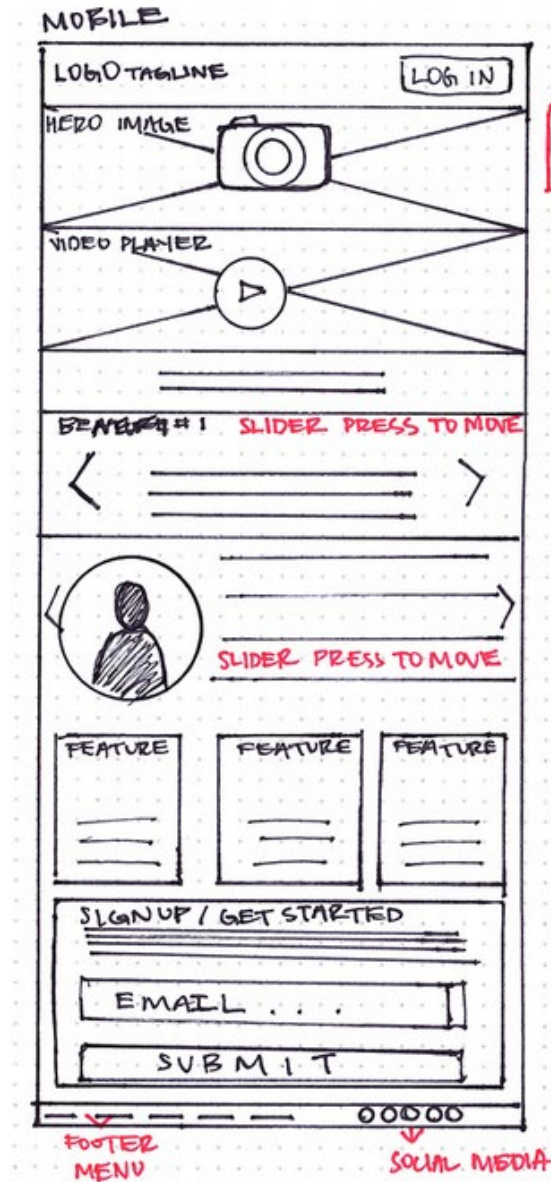
July 2002 survey of usability practitioners  
172 responses  
Numbers do not sum to 100% due to rounding



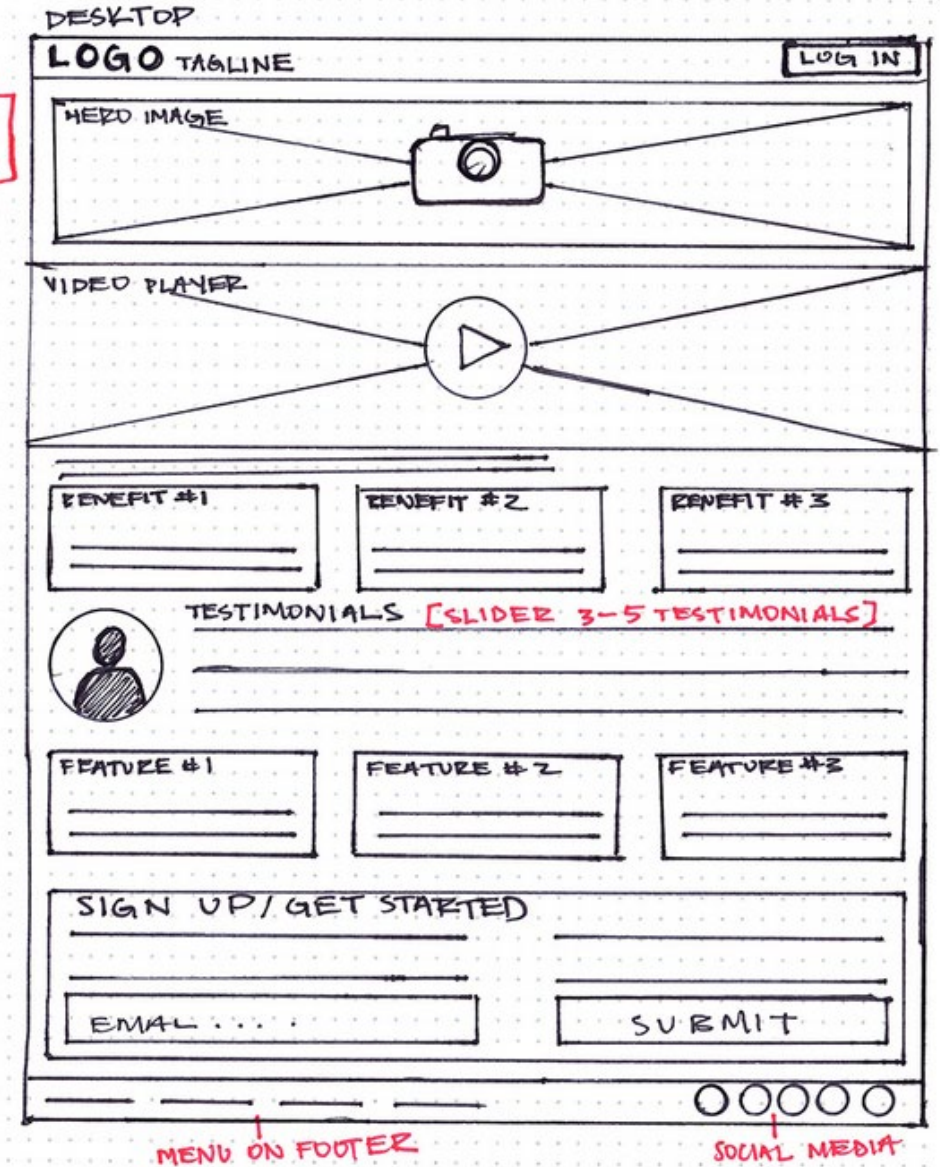
**Figure 1.12** Answers to the question, "What is the importance of paper prototyping to your work?" from a July 2002 survey of usability professionals.



# Low Fidelity: Wireframing



HOME  
PAGE



# Low vs High Fidelity

## Low Fidelity

- E.g., sketches/'Paper' prototypes.
- Focused on underlying ideas.
- Key functionality, content etc.
- Produced quickly.
- **Thrown away.**
- Generates many possible ideas.
- **Help client acceptance.**

## High Fidelity

- Built in software for automation.
- Similar style to final product.
- Accurate detail is important.
- Finalize chosen ideas.
- Still thrown away.
- Used in realistic studies.
- Helps client acceptance.



# Low vs High Fidelity

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- Low-fidelity – it captures the point, the functions etc.
  - To help improve the ideas.
- High-fidelity – represents **part** of the reality.
  - To agree on the final designs.
  - E.g., the finalised look and feel.
  - E.g., the finalised functionality.

# Is This High or Low?

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# What Are You Prototyping?

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# Defining Prototype Goals

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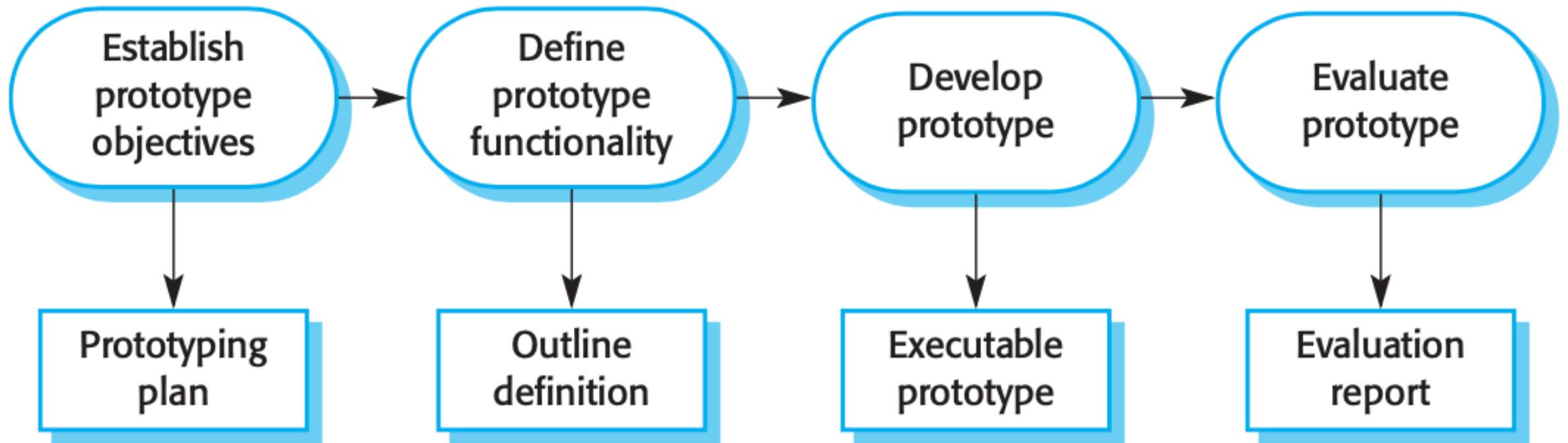
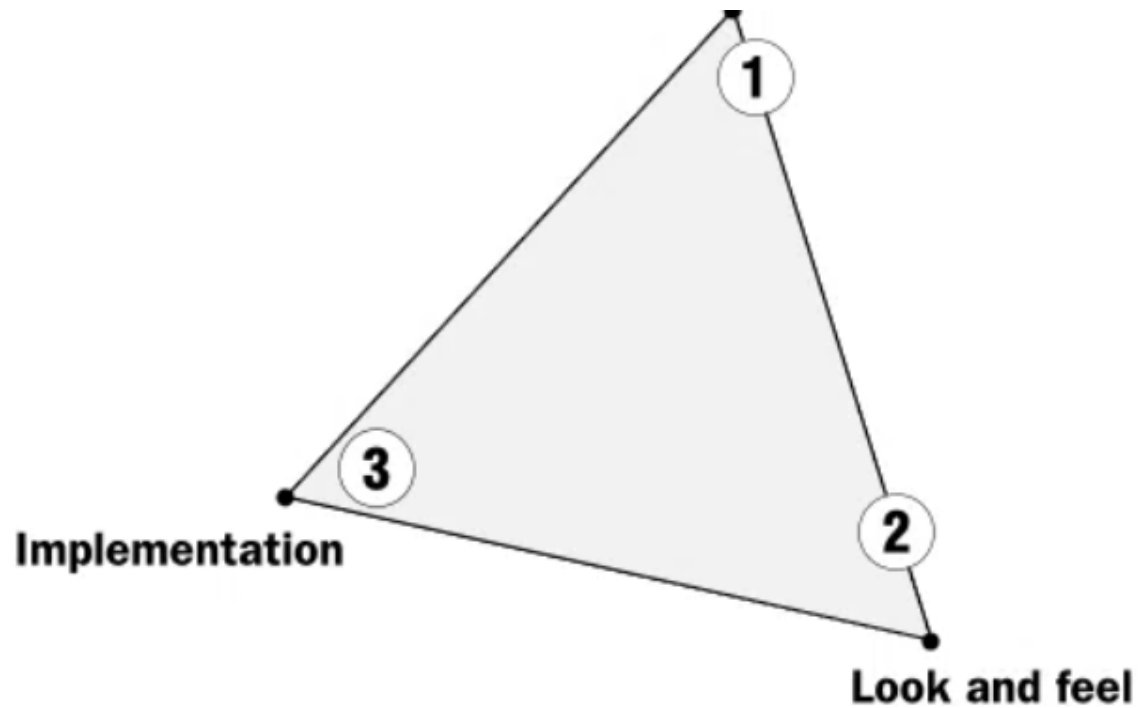


Fig: Prototype Development



# 3 Purposes of Prototype



- Prototypes fill a **purpose**.
  1. Role of technology.
  2. Look and Feel.
  3. Implementation Guide.
- UI Designers do #2.
- Powerpoint or paper prototypes do more #1.
- Apple Knowledge Navigator.
  - Aimed at all employees (company vision).
  - Does #1 – Does Not Do #2 or #3.

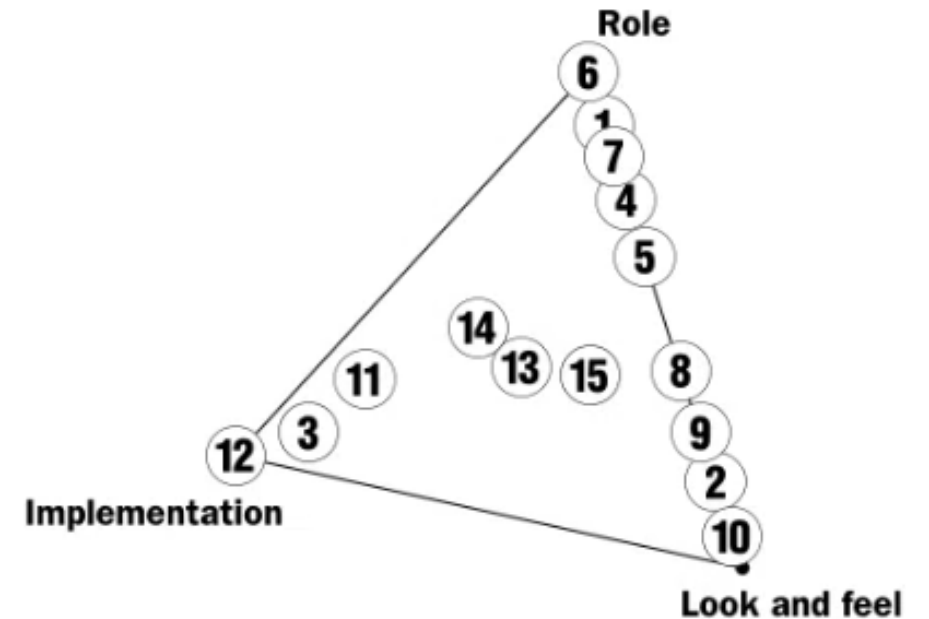
# Apple Aren't the Only Ones

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- Nokia 2006: <https://www.youtube.com/watch?v=-oPJTzPoc4>
- Nokia 2009: <https://www.youtube.com/watch?v=A4pDf7m2UPE>
- Nokia 2015: <https://www.youtube.com/watch?v=sXbPxDBzo7k>
  - Includes several things
    - Personas
    - Scenarios
    - Prototype concepts.
- Microsoft: <https://www.youtube.com/watch?v=w-tFdreZB94>

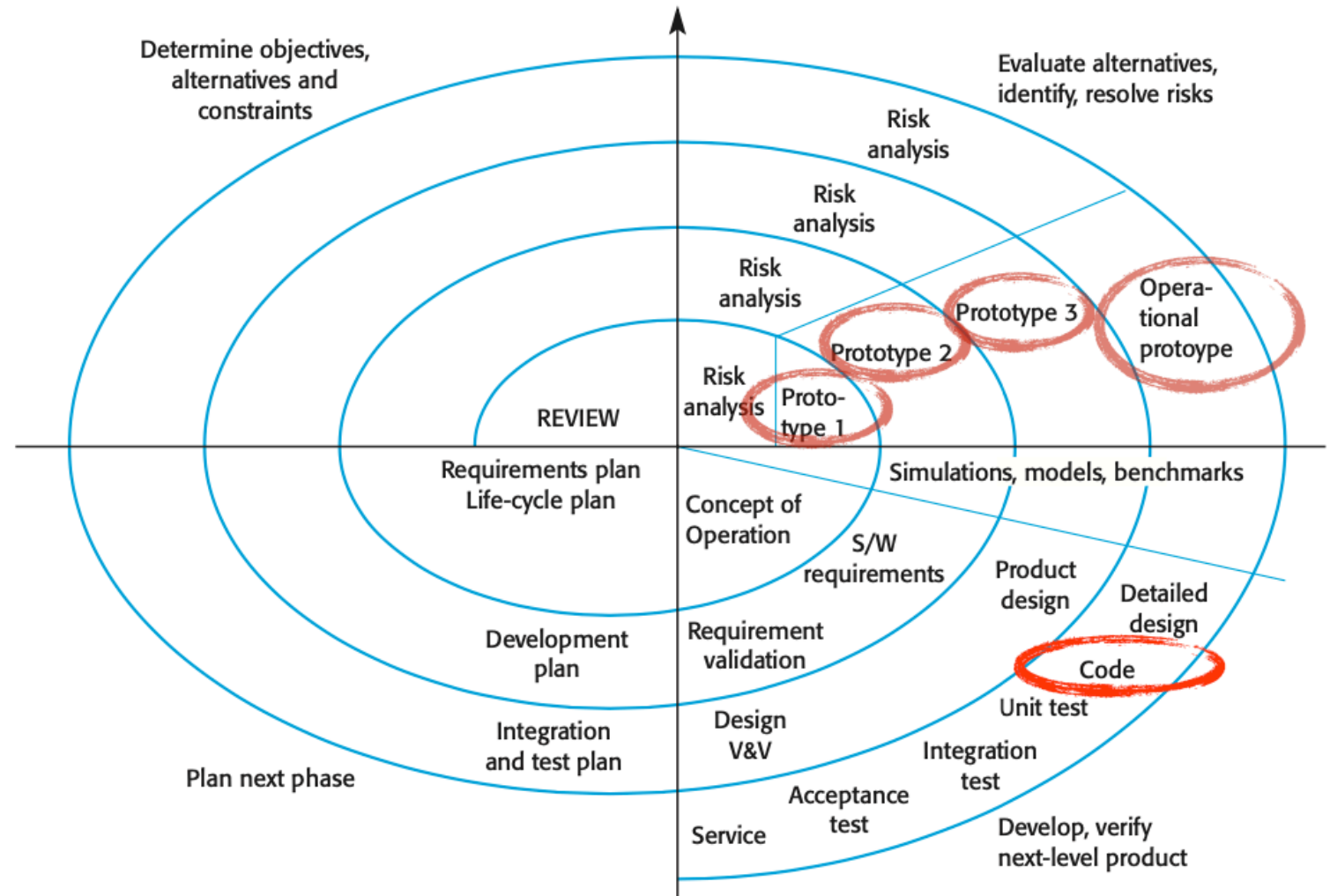
# Prototypes >> Implementation

- Prototypes help everyone to **image what we are building**.
  - The customer, the manager, the developer etc.
- But we still need to be able to tell a developer.
  - You go and build X.
  - Look & Feel does it for UI Devs.
  - Powerpoints could guide Implementation.
- Sometimes we **code up prototypes**.
  - These should be partial implementations.



1. 3D space-planning (role)
2. 3D space-planning (look and feel)
3. 3D space-planning (implementation)
4. Storyboard for portable notebook computer
5. Interactive story, operating system user interface
6. Vision video, notebook computer
7. Appearance model, integrated communicator
8. Animation, fashion design workspace
9. Look and feel simulation, child's toy
10. Pizza-box, architect's computer
11. Working prototype, digital movie editor
12. C++ program listing, fluid dynamics simulation
13. Integrated prototype, sound browser
14. Integrated prototype, pile metaphor
15. Integrated prototype, garment history browser

# Prototypes >> Implementations





# 10 Types of Prototypes (With Explanations and Tips)

Sketches and Diagrams

Wireframe

3D Printing Prototype

Physical Model

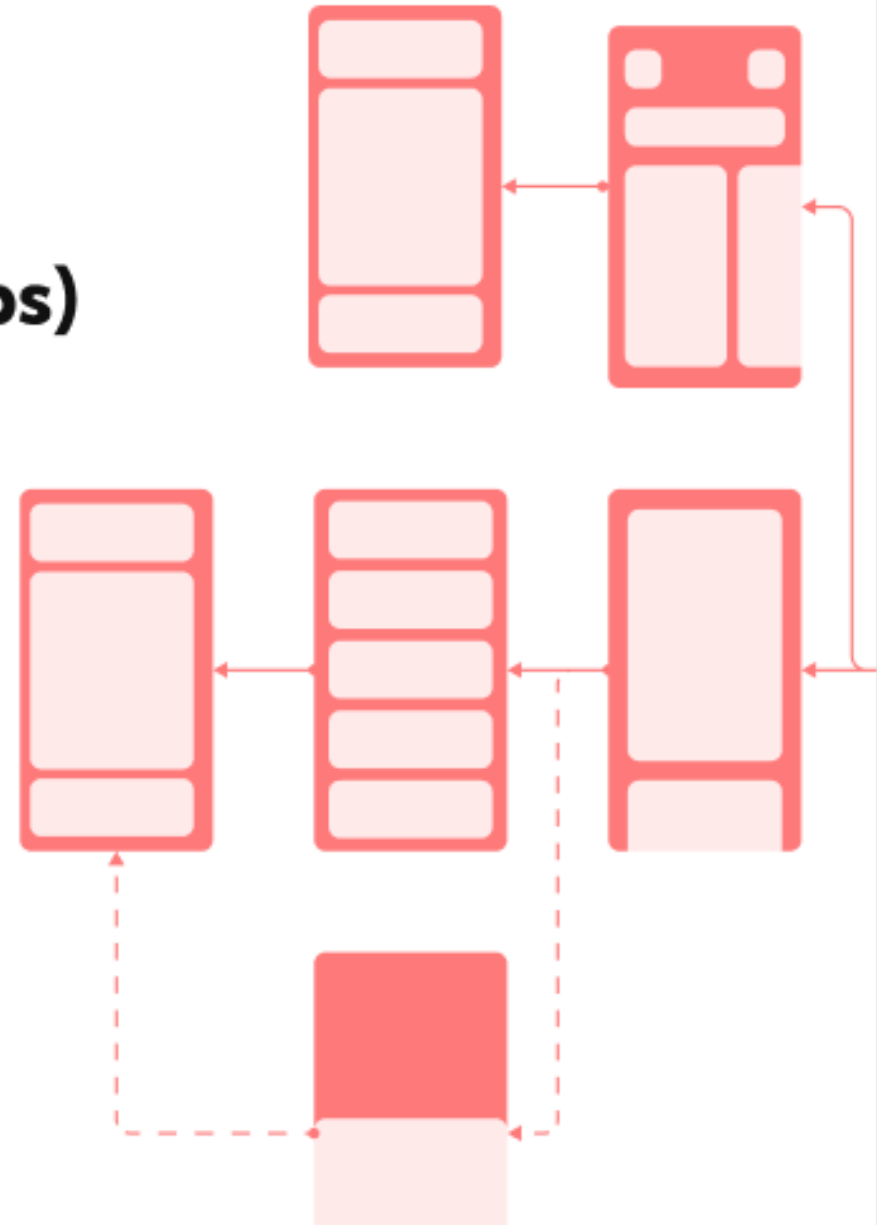
VR & AR Prototypes

Feasibility

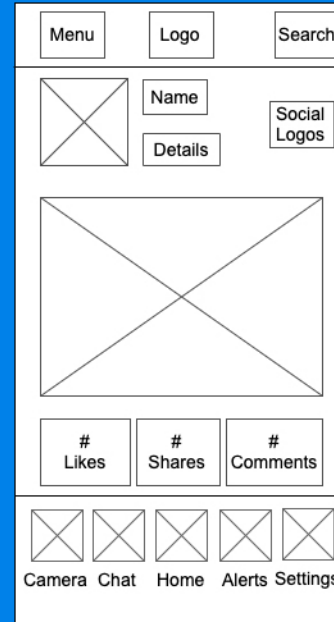
Video Prototype

Horizontal Prototype

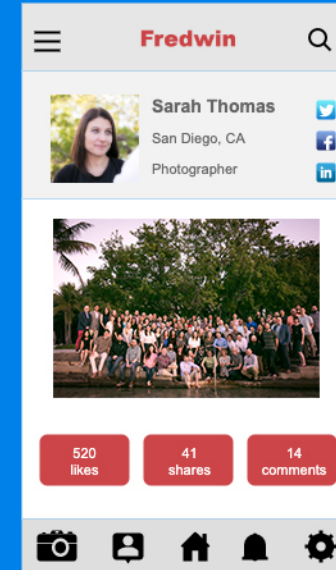
Fully Functional Rapid Prototype



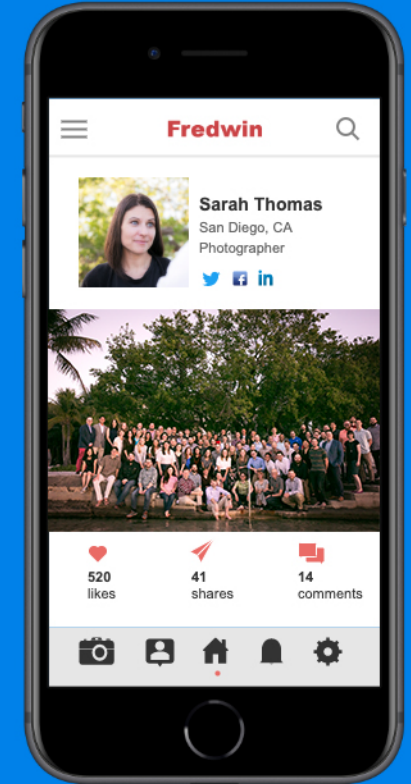
# Wireframe vs Mockup vs Code Prototype



Wireframe



Mockup



Prototype

# Wireframe vs Mockup vs Code Prototype

|                   | Wireframe                                                                           | Mockup                                                                                            | Code Prototype                                             |
|-------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| What              | A <b>quick sketch</b> to convey the high-level concept of new product functionality | A realistic <b>visual design</b> that resembles what the new product functionality will look like | <b>Interactive simulation</b> of new product functionality |
| Purpose           | To gain consensus and collect internal feedback on how new functionality will work  | To facilitate more detailed critiques of visual elements and functionality so changes can be made | To collect feedback by user testing the real experience    |
| Design fidelity   | Low                                                                                 | Middle                                                                                            | High                                                       |
| Included elements | The format and structure of content                                                 | Additional visual elements like logos, colors, and icons                                          | Final interactive elements and navigation                  |
| Time invested     | Low                                                                                 | Medium                                                                                            | High                                                       |
| Creator           | PM or UX Designer                                                                   | UX Designer                                                                                       | UX Designer                                                |

# Prototype vs Model

|         | Prototype                                                                               | Model                                                                                         |
|---------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| What    | A <b>preliminary version of a product</b> or system created for testing and evaluation. | A <b>representation or simulation of a system</b> used to understand or predict its behavior. |
| How     | Typically, physical or functional in nature.                                            | Can be physical or virtual.                                                                   |
| Focus   | Focused on <b>testing and refining</b> a specific product or system.                    | Focused on <b>understanding and predicting the behavior</b> of a broader system or process.   |
| Purpose | Used to <b>identify and fix design issues</b> .                                         | Used to <b>explore different scenarios and make prediction</b> .                              |
| Format  | May not be fully functional or representative of the final product or system.           | Designed to accurately represent the system being modeled.                                    |
| Stage   | Created early in the development process.                                               | Created at any stage of the development process.                                              |
| Cost    | Can be expensive to produce.                                                            | Can be less expensive to produce than physical prototypes.                                    |

# Prototyping Risks

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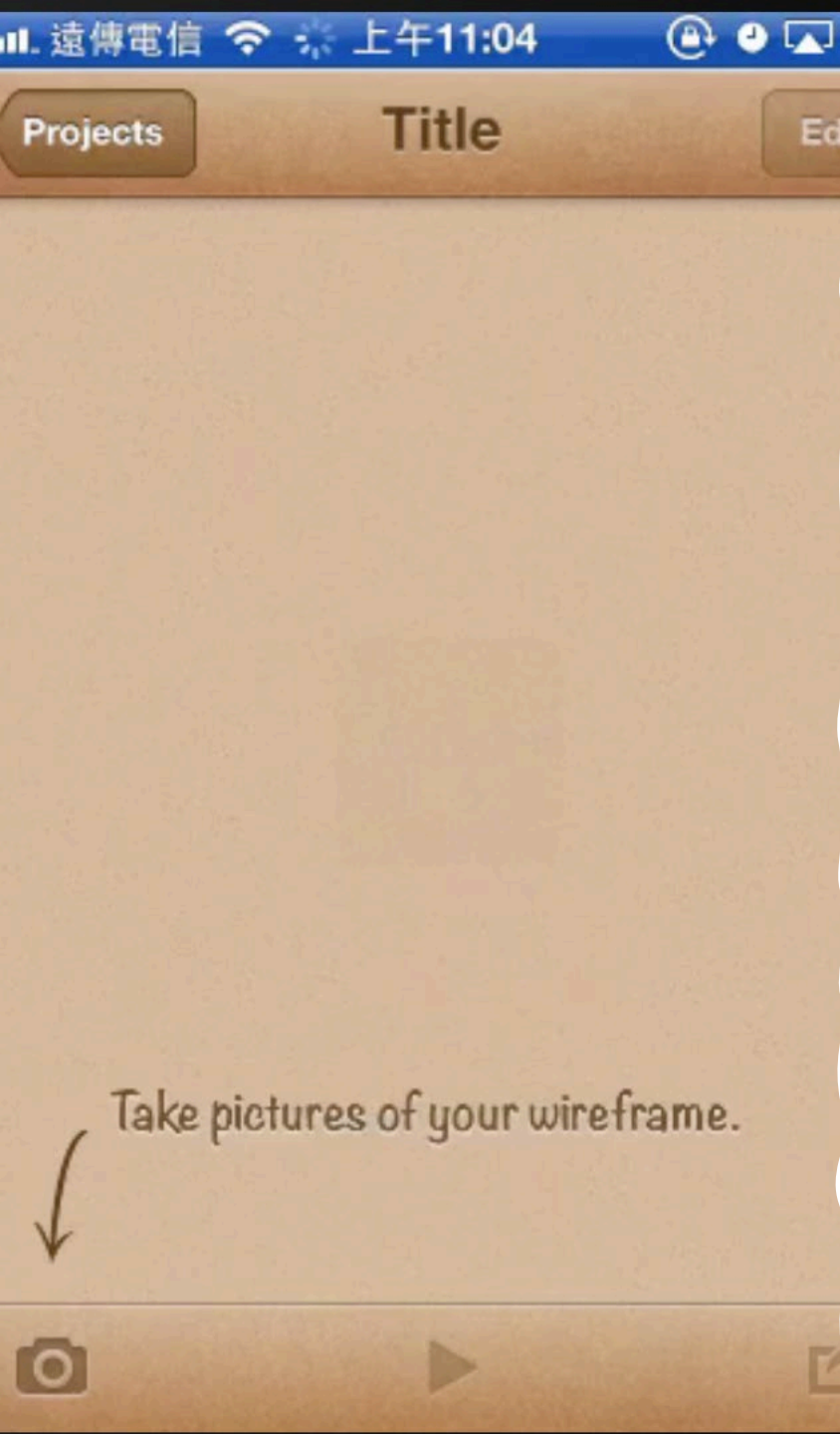
# Prototyping Risks

- 1. Investing too much time/energy on high-fidelity prototypes.**
  - When a low fidelity one would let you test the point.
2. Adhoc prototyping code is **re-used** in the real system.
  - But the code wasn't produced to a professional standard.
- 3. Prototyping is used instead of, rather than alongside, documentation.**
  - Which may then be insufficient for software maintenance phase.
4. Prototypes might be approved by the **wrong stakeholders**.
  - E.g., managers rather than the end users.



# Prototyping Tools

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# Prototyping Tools

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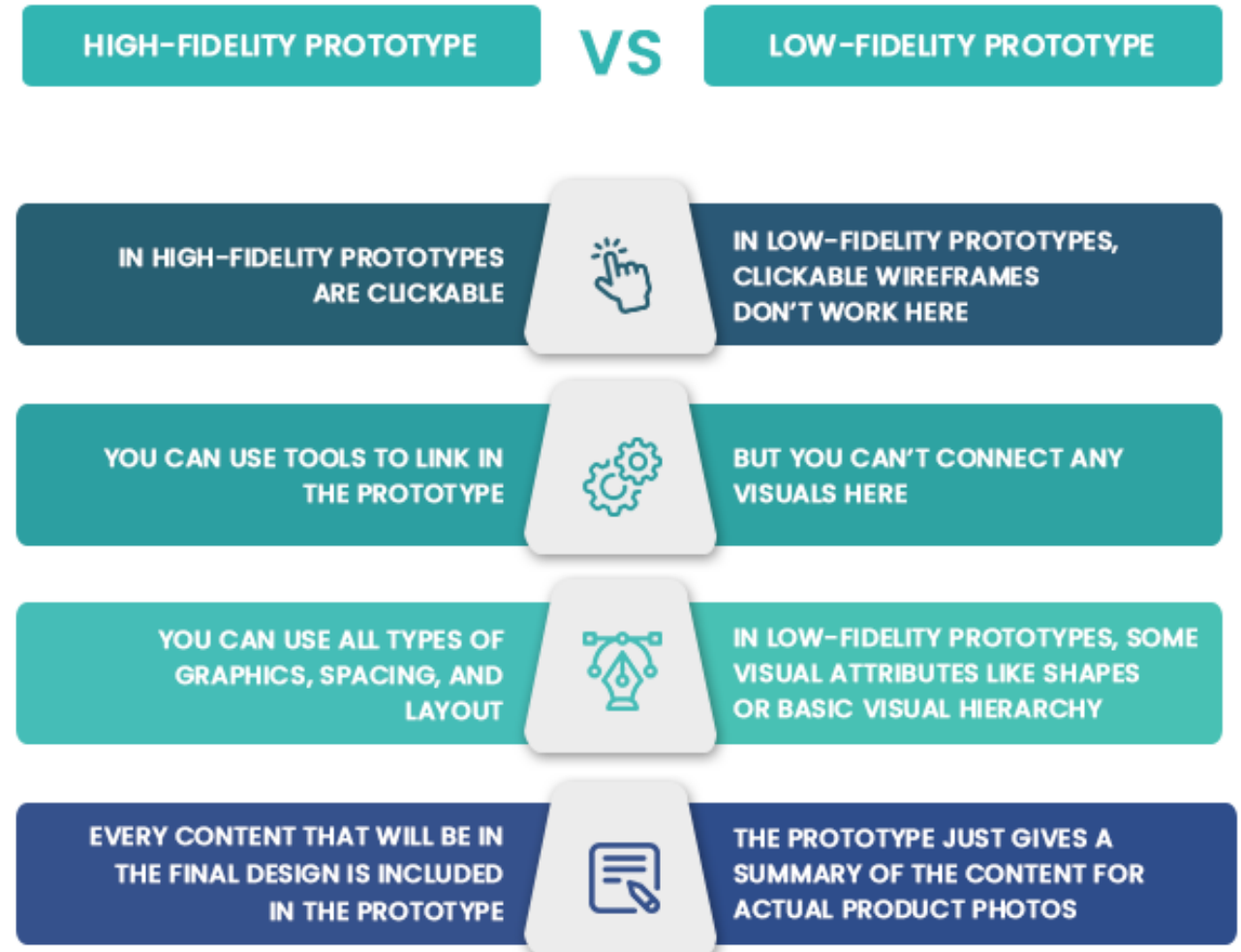
- Paper prototyping templates.
  - E.g., <https://sketchsheets.com/>
- Many good prototypes can be built in e.g., PowerPoint.
  - Can be semi-interactive!
  - Buttons link to different slides.
- Apps to help prototyping.
  - <https://marvelapp.com/>
  - <https://proto.io/en/demos/>
  - <https://prottapp.com/>
- Hi-Fi tools e.g., <https://www.sketch.com/>

# Summary

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- System Requirements -> Prototyping -> Implementation.
- More **time and efforts are focused on documentations** rather than coding.
- There are different members (roles) working at different phases of specifications, hence, it is important to understand that SE is targeting at a team collaboration with **different roles**.
- Understand the **purpose** of prototyping.

## HIGH-FIDELITY PROTOTYPE VS LOW-FIDELITY PROTOTYPE





THANK  
YOU